SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA07 **COURSE NAME: COMPUTER NETWORKS**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Kadam Mahesh(192312376) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mahesh

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X3+1.

   i.) Use polynomial long division to determine the message that should be transmitted.

   ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

SOLUTION:

Given:

Message: 11100011
Generator Polynomial: x
3

+
1

x3+1
Binary Generator: 1001

(i) CRC Calculation

Append 3 zeros (degree of polynomial = 3):

11100011 → 11100011000

Perform modulo-2 division by **1001**.

CRC Remainder = 100

Hence, the transmitted message is:

11100011100
11100011100

(ii) Error Detection

If the leftmost bit is inverted during transmission:

Transmitted: 11100011100
Received: 01100011100

Dividing the received message by **1001** gives:

Remainder = 001

Since the remainder is **not 000**, the receiver detects that an error has occurred.

Final Answer

CRC: 100
Transmitted message: 11100011100
Received remainder after error: 001
Conclusion: Non-zero remainder indicates a transmission error.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

SOLUTION:

(a) Throughput of the Network

Given:

Bandwidth = 10 Mbps
Frames transmitted = 12,000 frames/minute
Frame size = 10,000 bits

Formula:

Throughput = (Frames × Bits per frame) / Time

Calculation:

Throughput = (12,000 × 10,000) / 60

= 120,000,000 / 60

= 2,000,000 bps

= 2 Mbps

Answer: The throughput of the network is **2 Mbps**.

(b) Channel Capacity

Given:

Bandwidth (B) = 3000 Hz
Signal-to-Noise Ratio (S/N) = 3162

Formula (Shannon Capacity):

$C = B \times \log_2(1 + S/N)$

Calculation:

$C = 3000 \times \log_2(3163)$

$\approx 3000 \times 11.63$

$\approx 34{,}890$ bps

≈ **34.9 kbps**

Answer: The channel capacity is **34.9 kbps**.

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

SOLUTION:

Hamming (12,8) Code – Hospital Scenario

Problem:
A multispecialty hospital uses wireless patient-monitoring devices to continuously send patients' heart rate and blood pressure readings to a central monitoring station. Due to electromagnetic interference, a **single-bit error** may occur during data transmission. To ensure accurate data without retransmission, the hospital uses the **Hamming (12,8) error-correcting code**.

Solution:

The **Hamming (12,8)** code converts every **8-bit data** into a **12-bit codeword** by adding **4 parity bits**.
During transmission, if a **single bit is changed**, the parity bits help the receiver detect the exact position of the error.
The receiver automatically **corrects the erroneous bit** without asking the sender to retransmit the data.
This enables continuous and reliable monitoring of patients, even in the presence of wireless interference.

Advantages:

Detects and corrects **single-bit errors**.
Eliminates the need for retransmission.
Reduces communication delay.
Improves the reliability and accuracy of patient-monitoring data.

Ensures doctors receive correct patient information for timely medical decisions.

Conclusion:

By using the **Hamming (12,8) error-correcting code**, the hospital ensures that patient data is transmitted accurately and efficiently. Even if a single-bit error occurs due to electromagnetic interference, the receiver can detect and correct it automatically, providing reliable real-time healthcare monitoring.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

SOLUTION:

(a) IPv4 Subnetting

Given:

Network: **172.16.10.0/24**
Subnet Mask: **/27**

Step 1: Number of Subnets

Borrowed bits = 27 − 24 = 3

Number of subnets = 2^3 = **8**

Step 2: Hosts per Subnet

Host bits = 32 − 27 = 5

Total hosts = 2^5 = 32

Usable hosts = 32 − 2 = **30**

Step 3: Find the Subnet of 172.16.10.78

Subnet ranges:

172.16.10.0 – 31
172.16.10.32 – 63
172.16.10.64 – 95
172.16.10.96 – 127
172.16.10.128 – 159
172.16.10.160 – 191
172.16.10.192 – 223
172.16.10.224 – 255

IP **172.16.10.78** belongs to the subnet **172.16.10.64/27**.

Network Address = **172.16.10.64**

Broadcast Address = **172.16.10.95**

Usable Host Range = **172.16.10.65 – 172.16.10.94**

Step 4: Check IP 172.16.10.95

172.16.10.95 is the **broadcast address** of the subnet.

Therefore, it **does not represent a usable host**.

Answer:

Total Subnets = **8**
Usable Hosts/Subnet = **30**
Subnet = **172.16.10.64/27**
Network Address = **172.16.10.64**
Broadcast Address = **172.16.10.95**
IP 172.16.10.95 is the broadcast address, so it is **not a usable host**.

(b) IPv6 Addressing

Given Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed Form

2001:db8::ff00:42:8329

Expanded Form

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

Network Prefix:

2001:0db8:0000:0000::/64

Number of Host Addresses

Host bits = 128 − 64 = 64

Total Hosts = 2^{64}

= 18,446,744,073,709,551,616

Answer:

Compressed Address = **2001:db8::ff00:42:8329**
Expanded Address = **2001:0db8:0000:0000:0000:ff00:0042:8329**
Network Prefix = **2001:db8:0:0::/64**
Total Host Addresses = **2^{64} (18,446,744,073,709,551,616)**

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

SOLUTION:

Primary Networking Problem:
The primary networking problem is **Flow Control**. Flow control is used to regulate the rate at which the sender transmits data so that the receiver can process it without being overwhelmed.

Explanation:

- The university server is connected through a **10 Gbps** network and sends encrypted question papers to **8,000 students** simultaneously.
- Students use different internet connections, such as **100 Mbps fiber**, **20 Mbps Wi-Fi**, and **5–10 Mbps mobile networks**.
- Since many students have slower connections, they cannot receive and process data as quickly as the server transmits it.

- This mismatch between the sender's transmission speed and the receiver's processing capability causes the **receiver's buffer to overflow**.
- As a result, packets are lost, acknowledgments are delayed, and the sender retransmits the lost packets.
- These unnecessary retransmissions increase network traffic, leading to congestion, incomplete downloads, temporary freezing of the examination portal, and delays in delivering the question papers.

Solution:

- Flow control mechanisms, such as the **Sliding Window Protocol**, allow the receiver to inform the sender about the amount of data it can accept.
- The sender adjusts its transmission rate based on the receiver's available buffer space.
- This prevents buffer overflow, reduces packet loss and retransmissions, improves network efficiency, and ensures that all students receive the examination papers reliably and on time.

Conclusion:

The issue arises because the sender transmits data much faster than some receivers can handle. By implementing effective **flow control**, the network can balance the transmission rate with the receiver's processing speed, ensuring smooth, reliable, and efficient communication during the online examination.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandi n g	20%	Clearly interprets problem, identifies all parameters and requirement s accurately	Understan d s p r o b l e m with minor gaps	Partial understandi ng ; misses key elements	Misinterpre ts or unable to understand problem
Method Selection	20%	Selects most appropriate method/ formul a with strong justification	Selects correct method with minor errors	Inappropriat e or partially correct method	Unable to select method
Soluti on Proce ss	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear proce ss
Accuracy of Calculatio n	20%	All calculations accurate; correct final answer	Minor calculati on errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretati o n with minor omissions	Limited or unclear interpretati on	No interpretati o n